



KOHLER AI BATHROOM DESIGNER & PLANNER

AI-Powered Personalized Bathroom Planning





SYSTEM ARCHITECTURE:

1

User Input: Natural Language query or Guided Form : captures style, budget, space, and priorities.

2

LLM Extraction: Gemini parses intent into validated requirements : budget, style, priority, categories, dimensions.

3

Recommendation Engine: Hard constraints filter catalog; weighted scoring + Pareto pruning + Top-K selects best bundles.

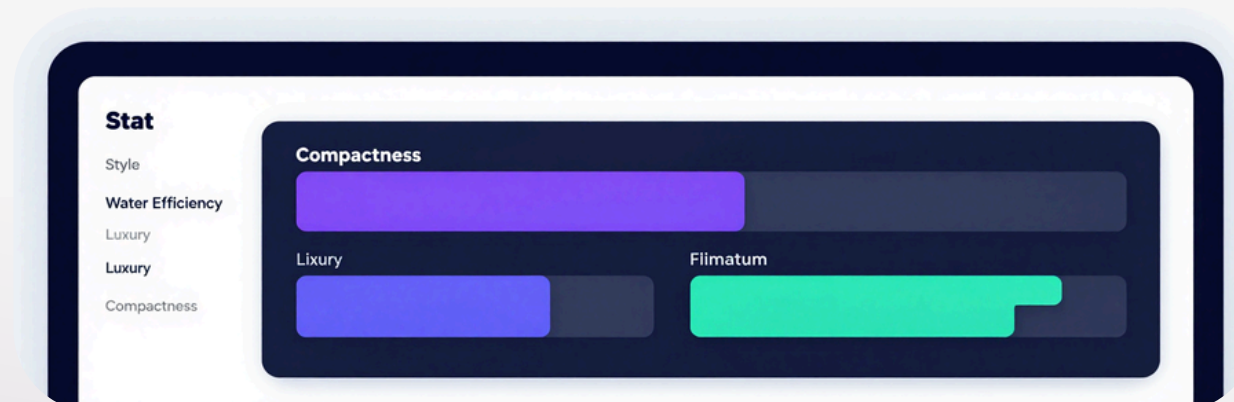
4

Output: Product bundle recommendations with factual explanation + Matplotlib 2D layout from catalog dimensions.



CORE APPROACH & TECH STACK

Hard constraints enforce feasibility;
weighted scoring adapts to user
priority across four dimensions:



1

Python + Pandas for Data
Processing

2

LLM API: gemini-2.0-flash-lite

3

Gradio Interactive UI

4

Matplotlib 2D Layout Viz

5

Constraint Filter -> Scoring -> Pareto
Pruning -> Top-K





FUTURE SCOPE & POSSIBILITIES

1

RAG Integration:

retrieve relevant information from KOHLER product manuals, specifications, installation guides, catalogs, and FAQs before generating recommendations.

2

Multimodal AI:

Understand bathroom images, floor plans, sketches, and natural-language requirements.

3

Agentic AI:

Automate the complete workflow from requirement extraction to product selection and layout generation.

4

Generative Design

Optimization:

Generate and evaluate multiple layouts based on budget, aesthetics, and spatial constraints.